

Environment High level

Description

This task is to optimize the operation of a cleaning robot moving through an area modeled as a 2D array. The ultimate goal of the robot is to complete its cleaning task, that is collecting garbage without any human intervention. It should not run out of battery midway through or get stuck.

- The map has garbage at random places and it also has places where the robot is likely to fall such as sand and grass (risky terrain types).
- The robot is placed at an arbitrary location on pavement.
- The grid has a singular charging station where a robot can charge its battery. The location of this charging station is another arbitrary location on pavement, where the agent does not spawn, symmetric to the trashcan.
- There is a bin marked on the grid that the robot can deposit the trash. The robot is capable of holding 3 pieces at a time.
- The map is a 13x13 2D numpy array that holds singular characters that represent the terrain type. Each reset() call bushes, charging station, the trash can and the agent are placed randomly.

State Space

1. Agent position
2. Current terrain type
3. Agent Trashload
4. Environment trash amount
5. Trash positions
6. Charging station position
7. Agent battery level
8. Trash bin position

Observation Space

1. The grid terrain types
2. It's location
3. Battery level
4. Trashload

Action Space

1. Move up
2. Move down
3. Move right
4. Move left
5. Pick up trash
6. Recharge
7. *Deposit Trash*

Rewards

1. Runs out of battery $\rightarrow -1$
2. Gets stuck (terrain) $\rightarrow -1$
3. Recharge $<80\% \rightarrow +0.05$
4. Recharge $>80\%$ and $<99\% \rightarrow +0.01$
5. Recharge $\geq 99\% \rightarrow -0.01$
6. Collect trash $\rightarrow +0.2$
7. *Drop trash in bin* $\rightarrow +0.5$
8. Attempts to pick up trash but fail, hold limit met or not on trash $\rightarrow -0.05$
9. Attempt to drop trash but not at trashcan $\rightarrow -0.05$

Start State

The agent is placed at an arbitrary pavement location that is not where the trashcan or charging station is.

Episode end

- Termination
 - Robot runs out of battery.
 - Robot gets stuck.
 - All trash collected and deposited
 - Truncation
 - 20,000 iterations (subject to change upon further testing)
-
- Rendering - Jackson
 - Map - Connor
 - Battery - jackson
 - Terrain types - Connor
 - Bush collision - Connor
 - Random garbage placement - Connor
 - Pick up litter / action space - jackson
 - Trash can and charging symmetric - Connor